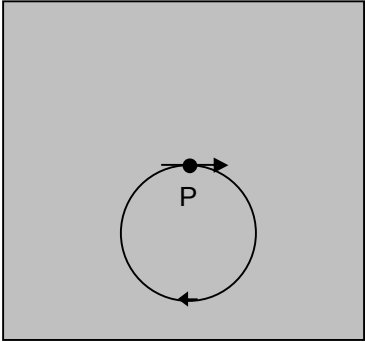


	not be able to be measured by the voltmeter.	
6(a)	Magnetic flux density is defined as the magnetic force per unit length per unit current acting on a straight conductor placed perpendicular to the magnetic field	[1] or zero
6(b)	Charged particle <u>is stationary</u> . Charged particle <u>is moving parallel to magnetic field</u> .	[1] [1]
6(c)(i)		[1] correct shape and direction.
6(c)(ii)	Since particle is moving in a circle, $F_B = F_C$ $Bqv = \frac{mv^2}{r}$ $2.0 \times 10^{-3} \times 1.60 \times 10^{-19} = \frac{9.11 \times 10^{-31} \times 2.5 \times 10^7}{r}$ $r = 0.071 \text{ m}$	[1] correct substitution [1] answer
6(c)(iii)	There will be an electric force on the electron directed out of the page. Electron will move in a circle due to the magnetic field and at the same time accelerating out of the page due to the electric field. The path of the electron will be a helix, the electron will travel a larger distance outwards per unit time due to its acceleration.	[1] explain effect of electric force on the electron [1] motion of electron

7(a)	mean power(D) I^2 D	[1] for
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